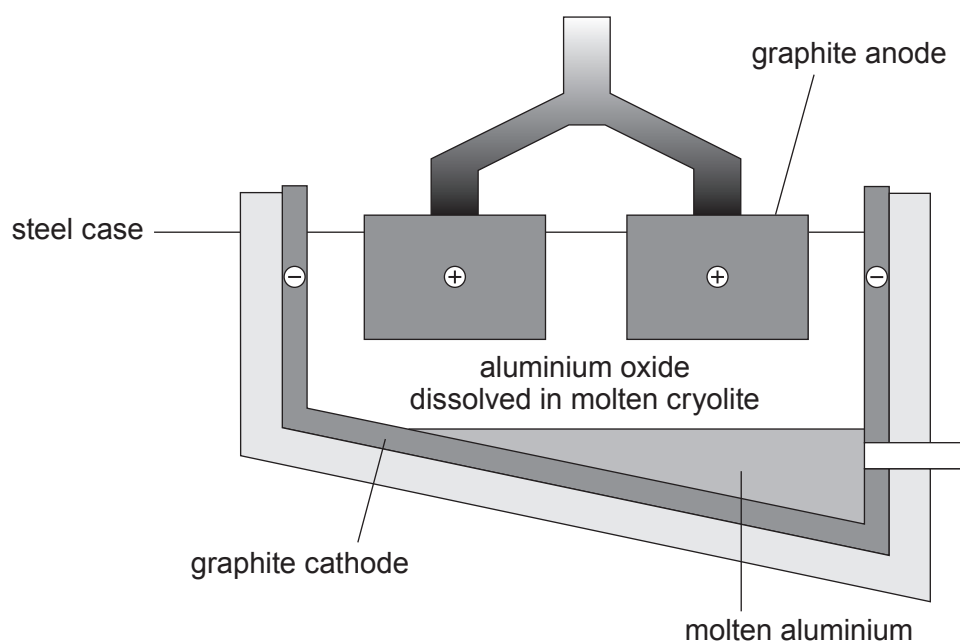


4. (a) Aluminium metal is produced from aluminium oxide in a process called electrolysis. The diagram shows a cell used in electrolysis.



- (i) Underline the correct word in the brackets to complete each sentence. [4]

Aluminium oxide is a (**compound** / mixture / element) of aluminium and oxygen.

Cryolite is added to (**raise** / lower / maintain) the temperature used for electrolysis.

Electrolysis uses a lot of (**electrical** / solar / chemical) energy.

Aluminium is produced as a (**solid** / liquid / gas).



(ii) Aluminium oxide contains the ions Al^{3+} and O^{2-} .

Give the formula of aluminium oxide. [1]

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(iii) Iron oxide is reduced by heating with carbon in a blast furnace.

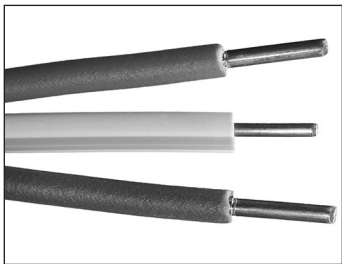
Tick (✓) the box that states why aluminium oxide cannot be reduced in this way. [1]

carbon is more reactive than aluminium	
iron is more reactive than aluminium	
aluminium is more reactive than carbon	



Examiner
only

(b) The pictures show copper metal used in electrical wiring, saucepans and water pipes.



Describe how the properties of copper make it suitable for each of these uses. [6 QER]

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